



Department of Industrial Engineering

IE48L Assignment 1

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Part 1. List of Interfaces

As we went through a typical day, we noted all the interfaces we encountered. Below is our comprehensive list of everyday interfaces:

- Card access system on campus gate
- Door handle
- Liquid medicine dispensers (measuring spoon, dropper, nasal spray)
- Light switch buttons
- Trash can lid opener (pedal)
- Hair dryer buttons and switches
- Computer keyboard
- Computer mouse
- USB-C connector
- Faucet handle (single-lever mixer)
- Automatic sliding door
- Revolving door
- Supermarket self-checkout kiosk
- Coffee machine buttons
- Elevator buttons and floor indicator
- Smartphone touchscreen and lock screen
- Car dashboard controls (speedometer, fuel gauge, turn signal stalk)
- Car air conditioning and climate control panel

Part 2. Good and Bad Interfaces

Bad Interfaces

1. Revolving Door

Revolving doors present a frustrating interaction for many users. The pacing is dictated by the mechanism rather than the user, creating an uncomfortable experience. There is poor feedback about when to enter or exit the compartment, and the mapping between pushing force and rotation speed is often unclear. As Norman discusses in "The Design of Everyday Things," a well-designed object should have visible affordances that signal how it should be used. Revolving doors frequently lack clear signifiers: users hesitate at the entrance, unsure which compartment to step into, how fast to walk, or how hard to push. The gulf of execution is wide — the user must figure out the correct timing and force, which is especially problematic for elderly people, children, or anyone carrying luggage. Additionally, the constraints are physical and unforgiving: getting caught between panels is a genuine safety concern.

2. Car Air Conditioning Controls

Many car climate control systems suffer from poor **mapping** and a frustrating lack of immediate **feedback**. A common issue is the temperature dial; instead of using precise **signifiers** like numeric values, many designs rely on a vague blue-to-red gradient. This ambiguity forces drivers into a "trial-and-error" process, widening the **Gulf of Execution**—the interface doesn't clearly show how much to turn the dial to reach a specific comfort level. The problem is worsened by a "feedback gap." There is a physical delay between turning the knob and actually feeling the air temperature change. Without instant confirmation, the user is left in the **Gulf of Evaluation**, unable to determine if their adjustment was sufficient. As Norman explains in *the Design of Everyday Things*, when feedback isn't immediate, users tend to overcorrect. In a driving context, this constant fidgeting increases cognitive load and creates a dangerous distraction.

3. Single-Lever Faucet Handle

The single-lever mixer faucet attempts to control two variables — water temperature and flow rate — with a single control, and does both poorly. Moving the lever up and down controls flow, while left-right controls temperature. The mapping is not natural, and the sensitivity is non-linear: a small movement near the center can cause a dramatic temperature swing, while larger movements at the extremes barely change anything. There is no tactile feedback (clicks or detents) to indicate where "warm" ends and "hot" begins. Norman describes this as a violation of good mapping: the relationship between the control and its effect should be intuitive and proportional. Users are forced to make constant micro-adjustments, often oscillating between too hot and too cold — a textbook example of poor feedback and poor conceptual model.

Good Interfaces

1. Hair Dryer Buttons and Switches

Most hair dryers feature a simple, well-mapped interface: a rocker switch for heat level (low/medium/high) and a separate button for fan speed. The controls have clear affordances — the physical shape of the switch tells you it slides, and distinct click positions provide tactile feedback for each setting. This is consistent with what Norman calls a "good conceptual model": users immediately understand that one switch controls heat and the other controls airflow. The system also satisfies the discoverability principle, as all controls are visible and accessible. As noted in "Three Teapots," a design that is both functional and easy to use builds trust and satisfaction.

2. The USB-C Connector

The USB-C connector is a perfect example of how thoughtful design can completely eliminate a common user's frustration. Before USB-C, older standards like Micro-USB or Type-A were notorious for being one-way only. We've all experienced that annoying "trial-and-error" loop, trying to plug a cable in, failing, flipping it, and sometimes failing again. By making the connector entirely symmetrical, USB-C ensures there is no wrong way to plug it in.

From a Norman-inspired perspective, this design removes a major **physical constraint** that used to slow us down. It bridges the **Gulf of Execution** instantly because you no longer need to squint at the cable or the port before making a move; you just plug it in. Additionally, the satisfying tactile click you feel when it seats properly provides immediate **physical feedback**. Bu confirms a secure connection and satisfies the **Gulf of Evaluation**, letting you know the device is charging without even looking at the screen.

3. Automatic Sliding Door

Automatic sliding doors represent a near-ideal interface: zero learning curve, zero effort, and instant response. The user's intent (approaching the door) is the only input required. The system relies on a motion sensor that maps the natural action of walking toward a door to the desired result of the door opening. As Norman discusses in "Emotion & Design," positive emotional responses arise when technology works seamlessly and removes barriers. The automatic door achieves this perfectly — there are no buttons to find, no handles to pull, no timing to guess. The feedback is immediate and visible (the door slides open), and the system gracefully handles errors (the door stays open if someone is in the threshold). It is a compelling example of an interface that "just works."

Part 3. Interface Revision — Single-Lever Faucet Handle

We chose the single-lever faucet handle for revision because it is an interface everyone uses multiple times daily, yet its design causes consistent frustration. The core problem is that a single lever must control two independent variables (temperature and flow rate) across a continuous range, with no feedback to guide the user. Drawing on Norman's principles from "The Design of Everyday Things," we propose a redesign that introduces better mapping, tactile feedback, and a more natural conceptual model.

Major Problems:

- Non-linear temperature sensitivity: A tiny movement near the center causes a drastic temperature change, while large movements at the extremes have little effect. Users cannot predict the outcome of their action.
- No tactile feedback: The lever moves smoothly across its range with no indication of temperature zones, so users must rely entirely on the delayed sensation of water temperature on their skin.
- Two-dimensional control with one lever: Combining vertical (flow) and horizontal (temperature) movement creates a complex mapping that violates the principle of natural mapping.
- Exhausting full-range adjustment: Going from minimum to maximum flow or temperature requires a sweeping arm motion that feels disproportionate to the task.

Proposed Solutions:

- Graduated resistance zones: Introduce mechanical detents (click stops) in the temperature axis at key points cold, warm, and hot. Between these zones, the lever should move freely for fine-tuning, but the clicks provide a "you are here" signal, much like the raised dot on the F key of a keyboard.
- Non-linear mechanical response (sensitivity curve): Redesign the internal cartridge so that temperature changes are proportional to lever displacement across the full range. Near the center (warm zone), the lever should require more movement per degree of change, allowing finer control in the most-used temperature range.
- Separate flow control: Add a collar or ring at the base of the lever that independently controls flow rate, separating the two variables. This creates a one-to-one mapping each control does exactly one thing.
- Visual temperature indicator: Integrate a thermochromic strip or small LED ring at the base of the faucet that changes color based on water temperature (blue → green → red), providing immediate visual feedback before the user touches the water.
- Reduced-travel mechanism: Use a gear ratio inside the cartridge that allows the full temperature range to be covered with a shorter lever travel, reducing the physical effort of adjustment.

These changes address Norman's fundamental design principles: the click stops provide constraints and feedback, the non-linear cartridge improves the mapping between action and outcome, the separate flow ring creates a better conceptual model (one control = one function), and the color indicator bridges the gulf of evaluation by making the system state visible. Together, they transform the faucet from a frustrating guessing game into an interface that communicates clearly with the user.